1. The algorithm for multiplication studied in school, known as long multiplication, consists of multiplying each digit in the first number by each digit in the second, and adding the results. What is the time complexity of long multiplication applied to two inputs, each with n digits?

2. Let $f \in C^\infty([a,b],\mathbb{R})$. Assume that Newton's method from a fixed initial guess $x_0 \in [a,b]$ gives a sequence of iterates which remain in $[a,b]$ and converges to $\alpha \in [a,b]$.
   (a) Show that Newton's method is at least quadratically convergent to α if $f'(\alpha) \neq 0$.
   (b) Show that Newton's method is linearly convergent to α if $f'(\alpha) = 0, f''(\alpha) \neq 0$. What is the rate of convergence in this case?
   (c) How does the rate of convergence vary if $f'(\alpha) = 0, f''(\alpha) = 0, f'''(\alpha) \neq 0$?
   (d) What is the order of convergence of Newton's method if $f'(\alpha) \neq 0, f''(\alpha) = 0$?

3. Apply Newton's method with $x_0 = 1/2$ to find the root $\alpha = 0$ of the following functions:
   (a) $f(x) = \sin x \sinh x$
   (b) $f(x) = x - \sin x$
   (c) $f(x) = x \cos x$

   Terminate the iteration when $|f(x)| < 10^{-8}$. In each case, relate the order of convergence observed to the previous question.

1) n digits multiply n digits so n^2 complexity for this part of method. Adding n numbers of up to $2n$ digits after multiplying has order n^2. Hence, overall the method has quadratic time complexity.

2)a) Let $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$. By the Taylor expansion, we obtain

$$f(x_i) = f(\alpha) + (x_i - \alpha)f'(\alpha) + \tfrac{1}{2}(x_i-\alpha)^2 f''(\alpha) + \tfrac{1}{6}(x_i-\alpha)^3 f'''(\alpha) + O(x_i^4)$$

$$f'(x_i) = f'(\alpha) + (x_i-\alpha)f''(\alpha) + \tfrac{1}{2}(x_i-\alpha)^2 \quad O(x_i^3)$$ and thus if $f'(\alpha) \neq 0$,

$$\frac{f(x_i)}{f'(x_i)} = \frac{(x_i-\alpha)f'(\alpha) + \frac{1}{2}(x_i-\alpha)^2 f''(\alpha) + \frac{1}{6}(x_i-\alpha)^3 f'''(\alpha) + O(x_i^4)}{f'(\alpha) + (x_i-\alpha)f''(\alpha) + \frac{1}{2}(x_i-\alpha)^2 f'''(\alpha) + O(x_i^3)}$$

$$= (x_i-\alpha) + \frac{f''(\alpha)}{2f'(\alpha)}(x_i-\alpha)^2 + O(x_i^3)$$

$x_{i+1} = \alpha + O(x_i^2)$ and so convergence is quadratic if $f'(\alpha) \neq 0$

b) If $f'(\alpha) = 0$, $f''(\alpha) \neq 0$, then $\frac{f(x_i)}{f'(x_i)} = \frac{1}{2}(x_i - \alpha) + O(x_i^2)$ and so linear convergence as $x_{i+1} = \frac{1}{2}(x_i + \alpha) + O(x_i^2)$. Rate is $1/2$.

c) If $f'(\alpha) = f''(\alpha) = 0$, $f'''(\alpha) \neq 0$, then $\frac{f(x_i)}{f'(x_i)} = \frac{1}{3}(x_i - \alpha) + O(x_i^2)$ and so $x_{i+1} = \frac{1}{3}(2x_i + \alpha) + O(x_i^2)$ so linear convergence with rate of $2/3$.

d) If $f'(\alpha) \neq 0$, $f''(\alpha) = 0$, then $\frac{f(x_i)}{f'(x_i)} = (x_i - \alpha) + \frac{f'''(\alpha)}{3f'(\alpha)}(x_i - \alpha)^3 + O(x_i^4)$ and thus $x_i = \alpha + O(x_i^3)$ so convergence is cubic.